

BRAC University

Department of Mathematics and Physical Sciences

Lecture 14

Pair of Straight Lines and Circles

Conditions, Equations, Intersections, Angles,
and Angle Bisectors

MAT110

Differential Calculus and Coordinate Geometry

Practice-Sheet-Based Lecture (Practice Sheet #9)

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1 General Equation of the Second Degree

The general equation of the second degree is

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0. \quad (1)$$

It represents a pair of straight lines when the left side can be factored as

$$(l_1x + m_1y + n_1)(l_2x + m_2y + n_2) = 0.$$

2 Condition for a Pair of Straight Lines

We regard (1) as a quadratic equation in x :

$$ax^2 + 2(hy + g)x + (by^2 + 2fy + c) = 0. \quad (2)$$

Assuming $a \neq 0$, the quadratic formula gives

$$x = \frac{-2(hy + g) \pm \sqrt{\{2(hy + g)\}^2 - 4a(by^2 + 2fy + c)}}{2a} \quad (3)$$

$$= \frac{-(hy + g) \pm \sqrt{(hy + g)^2 - a(by^2 + 2fy + c)}}{a}. \quad (4)$$

The expression under the square root is

$$(hy + g)^2 - a(by^2 + 2fy + c) = (h^2 - ab)y^2 + 2(hg - af)y + (g^2 - ac). \quad (5)$$

For each root in (4) to be linear in y , the quadratic expression in (5) must be a perfect square.

For $Ay^2 + 2By + C$ to be a perfect square, its discriminant must vanish:

$$\Delta_y = (2B)^2 - 4AC = 0 \iff B^2 - AC = 0.$$

Here

$$A = h^2 - ab, \quad B = hg - af, \quad C = g^2 - ac.$$

Therefore

$$(hg - af)^2 - (h^2 - ab)(g^2 - ac) = 0. \quad (6)$$

Expanding,

$$\begin{aligned} 0 &= h^2g^2 - 2afgh + a^2f^2 - (h^2g^2 - ach^2 - abg^2 + a^2bc) \\ &= -2afgh + a^2f^2 + ach^2 + abg^2 - a^2bc. \end{aligned}$$

Dividing by a and rearranging gives

$$\boxed{abc + 2fgh - af^2 - bg^2 - ch^2 = 0.}$$

Condition for a pair of straight lines

The general second-degree equation represents a degenerate pair of straight lines when

$$\begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} = 0 \iff abc + 2fgh - af^2 - bg^2 - ch^2 = 0.$$

If $a = 0$ but $b \neq 0$, the same proof follows by treating the equation as a quadratic in y .

Indeed, if the radicand is $(py + q)^2$, the two roots give the straight lines

$$ax + (h - p)y + (g - q) = 0, \quad ax + (h + p)y + (g + q) = 0.$$

3 Bisectors of the Angles Between Two Lines

If

$$L_1 = a_1x + b_1y + c_1 = 0, \quad L_2 = a_2x + b_2y + c_2 = 0,$$

then points on an angle bisector are equidistant from the two lines.

Angle-bisector formula

The two angle bisectors are

$$\frac{a_1x + b_1y + c_1}{\sqrt{a_1^2 + b_1^2}} = \pm \frac{a_2x + b_2y + c_2}{\sqrt{a_2^2 + b_2^2}}.$$

4 Examples

4.1

Show that

$$2y^2 - xy - x^2 + y + 2x - 1 = 0$$

represents a pair of straight lines. Find their point of intersection and the angle between them.

Comparing the equation with

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0,$$

we obtain

$$a = -1, \quad h = -\frac{1}{2}, \quad b = 2, \quad g = 1, \quad f = \frac{1}{2}, \quad c = -1.$$

Verification of the pair-of-lines condition. The required determinant is

$$D = \begin{vmatrix} -1 & -\frac{1}{2} & 1 \\ -\frac{1}{2} & 2 & \frac{1}{2} \\ 1 & \frac{1}{2} & -1 \end{vmatrix}.$$

Expanding D along the first row,

$$\begin{aligned} D &= (-1) \begin{vmatrix} 2 & \frac{1}{2} \\ \frac{1}{2} & -1 \end{vmatrix} - \left(-\frac{1}{2}\right) \begin{vmatrix} -\frac{1}{2} & \frac{1}{2} \\ 1 & -1 \end{vmatrix} + (1) \begin{vmatrix} -\frac{1}{2} & 2 \\ 1 & \frac{1}{2} \end{vmatrix} \\ &= (-1) \left(-2 - \frac{1}{4}\right) + \frac{1}{2} \left(\frac{1}{2} - \frac{1}{2}\right) + \left(-\frac{1}{4} - 2\right) \\ &= \frac{9}{4} + 0 - \frac{9}{4} = 0. \end{aligned}$$

Hence the given equation represents a pair of straight lines.

Determination of the two lines. Regard the complete given equation as a quadratic equation in x :

$$-x^2 + (2 - y)x + (2y^2 + y - 1) = 0.$$

Here

$$a = -1, \quad b = 2 - y, \quad c = 2y^2 + y - 1.$$

By the quadratic formula,

$$\begin{aligned} x &= \frac{-(2 - y) \pm \sqrt{(2 - y)^2 - 4(-1)(2y^2 + y - 1)}}{2(-1)} \\ &= \frac{y - 2 \pm \sqrt{4 - 4y + y^2 + 8y^2 + 4y - 4}}{-2} \\ &= \frac{y - 2 \pm \sqrt{9y^2}}{-2} = \frac{y - 2 \pm 3y}{-2}. \end{aligned}$$

Therefore the two roots are

$$x = \frac{y - 2 + 3y}{-2} = 1 - 2y, \quad \text{or} \quad x = \frac{y - 2 - 3y}{-2} = y + 1.$$

Thus the complete equation may be written as

$$2y^2 - xy - x^2 + y + 2x - 1 = (-x + y + 1)(x + 2y - 1).$$

Consequently, the equations of the two lines are

$$-x + y + 1 = 0, \tag{7}$$

$$x + 2y - 1 = 0. \tag{8}$$

Point of intersection. Adding (7) and (8) gives

$$3y = 0 \implies y = 0.$$

Substituting $y = 0$ into (8),

$$x - 1 = 0 \implies x = 1.$$

Therefore the point of intersection is

$$\boxed{(1, 0)}.$$

Angle between the lines. The slopes are $m_1 = 1$ and $m_2 = -\frac{1}{2}$. Hence

$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right| = \left| \frac{1 + 1/2}{1 - 1/2} \right| = 3.$$

Therefore the acute angle is

$$\theta = \tan^{-1}(3) \approx 71.565^\circ,$$

and the other angle is approximately 108.435° .

Equations of the angle bisectors. Using (7) and (8), the angle-bisector formula gives

$$\frac{-x+y+1}{\sqrt{(-1)^2+1^2}} = \pm \frac{x+2y-1}{\sqrt{1^2+2^2}}.$$

Hence

$$\frac{-x+y+1}{\sqrt{2}} = \pm \frac{x+2y-1}{\sqrt{5}}.$$

Therefore, the required bisectors are

$$\sqrt{5}(-x+y+1) - \sqrt{2}(x+2y-1) = 0 \quad (9)$$

and

$$\sqrt{5}(-x+y+1) + \sqrt{2}(x+2y-1) = 0. \quad (10)$$

4.2

Find k so that

$$2x^2 + xy - y^2 - 2x - 5y + k = 0$$

represents a pair of straight lines.

Comparing with the general equation gives

$$a = 2, \quad h = \frac{1}{2}, \quad b = -1, \quad g = -1, \quad f = -\frac{5}{2}, \quad c = k.$$

The determinant condition is

$$D = \begin{vmatrix} 2 & \frac{1}{2} & -1 \\ \frac{1}{2} & -1 & -\frac{5}{2} \\ -1 & -\frac{5}{2} & k \end{vmatrix} = 0.$$

Expanding along the first row,

$$\begin{aligned} D &= 2 \begin{vmatrix} -1 & -\frac{5}{2} \\ -\frac{5}{2} & k \end{vmatrix} - \frac{1}{2} \begin{vmatrix} \frac{1}{2} & -\frac{5}{2} \\ -1 & k \end{vmatrix} - \begin{vmatrix} \frac{1}{2} & -1 \\ -1 & -\frac{5}{2} \end{vmatrix} \\ &= 2 \left(-k - \frac{25}{4} \right) - \frac{1}{2} \left(\frac{k}{2} - \frac{5}{2} \right) - \left(-\frac{5}{4} - 1 \right) \\ &= -2k - \frac{25}{2} - \frac{k}{4} + \frac{5}{4} + \frac{9}{4} \\ &= -\frac{9}{4}k - 9. \end{aligned}$$

Therefore $D = 0$ implies

$$-\frac{9}{4}k - 9 = 0 \implies \boxed{k = -4}.$$

When $k = -4$, the equation becomes

$$2x^2 + xy - y^2 - 2x - 5y - 4 = 0.$$

Regard the complete equation as a quadratic equation in x :

$$2x^2 + (y - 2)x - (y^2 + 5y + 4) = 0.$$

Here

$$a = 2, \quad b = y - 2, \quad c = -(y^2 + 5y + 4).$$

Using the quadratic formula,

$$\begin{aligned} x &= \frac{-(y-2) \pm \sqrt{(y-2)^2 - 4(2)(-y^2-5y-4)}}{2(2)} \\ &= \frac{2-y \pm \sqrt{y^2-4y+4+8y^2+40y+32}}{4} \\ &= \frac{2-y \pm \sqrt{9y^2+36y+36}}{4} \\ &= \frac{2-y \pm 3(y+2)}{4}. \end{aligned}$$

Hence the two roots are

$$x = \frac{2-y+3(y+2)}{4} = \frac{y+4}{2}, \quad \text{or} \quad x = \frac{2-y-3(y+2)}{4} = -(y+1).$$

Thus

$$2x^2 + xy - y^2 - 2x - 5y - 4 = (2x - y - 4)(x + y + 1),$$

which confirms that the equation represents the two lines

$$2x - y - 4 = 0, \quad x + y + 1 = 0.$$

c

4.3

Find the angle bisectors of

$$8x^2 - 14xy + 6y^2 + 2x - y - 1 = 0.$$

Regard the complete given equation as a quadratic equation in x :

$$8x^2 + (2 - 14y)x + (6y^2 - y - 1) = 0.$$

Here

$$a = 8, \quad b = 2 - 14y, \quad c = 6y^2 - y - 1.$$

By the quadratic formula,

$$\begin{aligned}
 x &= \frac{-(2-14y) \pm \sqrt{(2-14y)^2 - 4(8)(6y^2 - y - 1)}}{2(8)} \\
 &= \frac{14y - 2 \pm \sqrt{4 - 56y + 196y^2 - 192y^2 + 32y + 32}}{16} \\
 &= \frac{14y - 2 \pm \sqrt{4y^2 - 24y + 36}}{16} \\
 &= \frac{14y - 2 \pm 2(y-3)}{16}.
 \end{aligned}$$

Hence the two roots are

$$x = \frac{14y - 2 + 2(y-3)}{16} = y - \frac{1}{2}, \quad \text{or} \quad x = \frac{14y - 2 - 2(y-3)}{16} = \frac{3y+1}{4}.$$

Consequently,

$$8x^2 - 14xy + 6y^2 + 2x - y - 1 = (4x - 3y - 1)(2x - 2y + 1).$$

Thus the equations of the lines are

$$4x - 3y - 1 = 0, \tag{11}$$

$$2x - 2y + 1 = 0. \tag{12}$$

Using (11) and (12), the angle bisectors satisfy

$$\frac{4x - 3y - 1}{5} = \pm \frac{2x - 2y + 1}{2\sqrt{2}}.$$

Therefore the required equations are

$$\boxed{2\sqrt{2}(4x - 3y - 1) - 5(2x - 2y + 1) = 0} \tag{13}$$

and

$$\boxed{2\sqrt{2}(4x - 3y - 1) + 5(2x - 2y + 1) = 0}. \tag{14}$$

4.4

Find the angle bisectors of

$$2x^2 + xy - y^2 - 3x + 6y - 9 = 0.$$

Regard the complete given equation as a quadratic equation in x :

$$2x^2 + (y-3)x - (y^2 - 6y + 9) = 0.$$

Here

$$a = 2, \quad b = y - 3, \quad c = -(y^2 - 6y + 9).$$

By the quadratic formula,

$$\begin{aligned}
 x &= \frac{-(y-3) \pm \sqrt{(y-3)^2 - 4(2)(-y^2 + 6y - 9)}}{2(2)} \\
 &= \frac{3-y \pm \sqrt{(y-3)^2 + 8(y^2 - 6y + 9)}}{4} \\
 &= \frac{3-y \pm \sqrt{9(y-3)^2}}{4} \\
 &= \frac{3-y \pm 3(y-3)}{4}.
 \end{aligned}$$

Thus the two roots are

$$x = \frac{3-y+3(y-3)}{4} = \frac{y-3}{2}, \quad \text{or} \quad x = \frac{3-y-3(y-3)}{4} = 3-y.$$

Therefore,

$$2x^2 + xy - y^2 - 3x + 6y - 9 = (2x - y + 3)(x + y - 3).$$

Hence the equations of the lines are

$$2x - y + 3 = 0, \tag{15}$$

$$x + y - 3 = 0. \tag{16}$$

Using (15) and (16), the angle bisectors satisfy

$$\frac{2x-y+3}{\sqrt{5}} = \pm \frac{x+y-3}{\sqrt{2}}.$$

Therefore

$$\boxed{\sqrt{2}(2x-y+3) - \sqrt{5}(x+y-3) = 0} \tag{17}$$

and

$$\boxed{\sqrt{2}(2x-y+3) + \sqrt{5}(x+y-3) = 0}. \tag{18}$$

5 Circle

5.1 Standard Form of a Circle

Circle with Given Centre and Radius

A circle with centre (α, β) and radius r has equation

$$(x - \alpha)^2 + (y - \beta)^2 = r^2. \quad (19)$$

Indeed, if $P(x, y)$ is any point on the circle, then its distance from the centre $C(\alpha, \beta)$ is r . Therefore,

$$CP = r \implies \sqrt{(x - \alpha)^2 + (y - \beta)^2} = r,$$

which gives (19) after squaring.

Circle Example 1

Find the equation of the circle with centre $(2, -3)$ and radius 4.

Here

$$\alpha = 2, \quad \beta = -3, \quad r = 4.$$

Using (19),

$$\begin{aligned} (x - \alpha)^2 + (y - \beta)^2 &= r^2 \\ (x - 2)^2 + (y - (-3))^2 &= 4^2. \end{aligned}$$

Hence the required equation is

$$(x - 2)^2 + (y + 3)^2 = 16.$$

In expanded form,

$$x^2 + y^2 - 4x + 6y - 3 = 0.$$

5.2 General Equation of a Circle

General Equation, Centre and Radius

The general equation of the second degree is

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0.$$

For this equation to represent a circle, the coefficients of x^2 and y^2 must be equal and the xy -term must be absent. After division by the common coefficient of x^2 and y^2 , the equation becomes

$$x^2 + y^2 + 2gx + 2fy + c = 0. \quad (20)$$

Completing the squares,

$$\begin{aligned}x^2 + 2gx + y^2 + 2fy &= -c, \\(x + g)^2 + (y + f)^2 &= g^2 + f^2 - c.\end{aligned}$$

Therefore,

$$\boxed{\text{Centre} = (-g, -f)}, \quad \boxed{\text{Radius} = \sqrt{g^2 + f^2 - c}}.$$

For a real circle, $g^2 + f^2 - c > 0$.

Circle Example 2

Find the centre and radius of

$$x^2 + y^2 - 6x + 4y - 12 = 0.$$

Comparing the given equation with (20),

$$2g = -6, \quad 2f = 4, \quad c = -12.$$

Thus

$$g = -3, \quad f = 2, \quad c = -12.$$

Hence the centre is

$$(-g, -f) = \boxed{(3, -2)},$$

and the radius is

$$\begin{aligned}r &= \sqrt{g^2 + f^2 - c} \\&= \sqrt{(-3)^2 + 2^2 - (-12)} = \sqrt{9 + 4 + 12} = \boxed{5}.\end{aligned}$$

Equivalently, completing the squares gives

$$\boxed{(x - 3)^2 + (y + 2)^2 = 25}.$$

5.3 Circle Passing Through Three Given Points

Circle Example 3

Find the equation of the circle passing through the points

$$(1, 3), \quad (-1, 1), \quad (2, -1).$$

Let the required circle be

$$x^2 + y^2 + 2gx + 2fy + c = 0. \quad (21)$$

Substituting the three given points into (21), we get

$$(1, 3): 1^2 + 3^2 + 2g + 6f + c = 0 \implies 2g + 6f + c = -10, \quad (22)$$

$$(-1, 1): (-1)^2 + 1^2 - 2g + 2f + c = 0 \implies -2g + 2f + c = -2, \quad (23)$$

$$(2, -1): 2^2 + (-1)^2 + 4g - 2f + c = 0 \implies 4g - 2f + c = -5. \quad (24)$$

Subtracting (23) from (22) and (24), respectively, gives

$$4g + 4f = -8 \implies g + f = -2, \quad (A)$$

$$6g - 4f = -3. \quad (B)$$

From (A), $f = -2 - g$. Substituting this into (B),

$$6g - 4(-2 - g) = -3,$$

$$10g + 8 = -3,$$

$$g = -\frac{11}{10}.$$

Therefore,

$$f = -2 - g = -2 + \frac{11}{10} = -\frac{9}{10}.$$

Using $g = -\frac{11}{10}$ and $f = -\frac{9}{10}$ in (23),

$$\begin{aligned} -2\left(-\frac{11}{10}\right) + 2\left(-\frac{9}{10}\right) + c &= -2, \\ \frac{22}{10} - \frac{18}{10} + c &= -2, \\ c &= -\frac{12}{5}. \end{aligned}$$

Therefore the required circle is

$$x^2 + y^2 - \frac{11}{5}x - \frac{9}{5}y - \frac{12}{5} = 0.$$

Multiplying by 5, we obtain

$$\boxed{5x^2 + 5y^2 - 11x - 9y - 12 = 0}. \quad (25)$$

6 Lecture Summary

Pair of Straight Lines

For $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$, the required condition is

$$\begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} = 0 \iff abc + 2fgh - af^2 - bg^2 - ch^2 = 0.$$

Treating the complete equation as a quadratic in x gives

$$x = \frac{-(hy + g) \pm \sqrt{(hy + g)^2 - a(by^2 + 2fy + c)}}{a}.$$

The two roots produce the two linear equations $L_1 = 0$ and $L_2 = 0$.

Circle

$$(x - \alpha)^2 + (y - \beta)^2 = r^2, \quad x^2 + y^2 + 2gx + 2fy + c = 0.$$

For the general equation, the centre is $(-g, -f)$ and the radius is $\sqrt{g^2 + f^2 - c}$.

Intersection, Angle and Angle Bisectors

For $L_1 = a_1x + b_1y + c_1 = 0$ and $L_2 = a_2x + b_2y + c_2 = 0$: solve the two equations simultaneously for their intersection, and use

$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|, \quad \frac{a_1x + b_1y + c_1}{\sqrt{a_1^2 + b_1^2}} = \pm \frac{a_2x + b_2y + c_2}{\sqrt{a_2^2 + b_2^2}}.$$

7 Practice Problems:

7.1 Pair of Straight Lines

- Show that the following equations represent pairs of straight lines. Find also their points of intersection and the angles between them:
 - $2y^2 - xy - x^2 + y + 2x - 1 = 0$;
 - $2x^2 - 2xy + x + 2y - 3 = 0$;
 - $x^2 + 3xy + 2y^2 + 18x - 321 = 0$;
 - $21x^2 + 40xy - 21y^2 + 44x + 122y - 17 = 0$.
- Find the values of λ or k so that the following equations may represent pairs of straight lines:
 - $2\lambda xy - y^2 + 4x + 2y + 8 = 0$;
 - $2x^2 + xy - y^2 - 2x - 5y + k = 0$;

(iii) $x^2 - \lambda xy + 2y^2 + 3x - 5y + 2 = 0$;

(iv) $12x^2 - 10xy + 2y^2 + 11x - 5y + \lambda = 0$.

3. Find the equations of the bisectors of the angles between the following pairs of straight lines:

(i) $x^2 + xy - 6y^2 - x - 8y - 2 = 0$;

(ii) $8x^2 - 14xy + 6y^2 + 2x - y - 1 = 0$;

(iii) $2x^2 + xy - y^2 - 3x + 6y - 9 = 0$;

(iv) $2x^2 + 7xy + 6y^2 + 13x + 22y + 20 = 0$.

7.2 Circle

1. Find the equation of the circle with:

(i) centre $(-2, -1)$ and radius 4;

(ii) centre $(9, 0)$ and radius 1;

(iii) centre $(0, 0)$ and radius 5.

2. Find the centre and radius of each of the following circles:

(i) $5x^2 + 5y^2 - 11x - 9y - 12 = 0$;

(ii) $x^2 + y^2 + 2x + 2y + 1 = 0$;

(iii) $x^2 + y^2 + 2x - 4y - 8 = 0$.

3. Find the equation of the circle passing through the following points:

(i) $(1, 3), (2, -1), (-1, 1)$;

(ii) $(-4, -3), (-1, -7), (0, 0)$;

(iii) $(3, 1), (4, -3), (1, -1)$.